

Rithvik Nakirikanti

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Technical Skills

Languages: Java, Python, Go, SQL, Bash

Backend: Spring Boot, FastAPI, Dropwizard, Kafka, REST APIs, Microservices, PostgreSQL, Redis

Frontend: React, TypeScript, Tailwind CSS, Vite

Infrastructure: Docker, Kubernetes, Terraform, Jenkins, Argo CD, CI/CD

Testing: JUnit, Testcontainers, Vitest, Playwright

Experience

Oracle Cloud Infrastructure | *Java, Dropwizard, OpenAPI, Kubernetes, Terraform*

Seattle, WA

Software Engineer (Member of Technical Staff)

Jul 2025 – Jun 2026

- Shipped PDB create and rename features on Oracle Database@Google, now used to provision and rename 500+ customer databases, by building the **Java/Dropwizard** endpoints and **OpenAPI** specs end-to-end.
- Cut Sev2 triage time by ~50% across 100+ weekly tickets by building an AI on-call assistant in **Codex** with **MCP** servers that pulls ticket context, service logs, and **Grafana** dashboards to draft incident recaps and severity assessments.
- Helped reduce new-region rollout time for Oracle Database@Google from ~2 weeks to ~1 week by adding regional **Terraform** config (endpoints, DNS zones) to existing modules for **OKE** cluster and control plane service provisioning.
- Eliminated broad service-account access on 50+ customer database operations in the Oracle Database@Google control plane by implementing scoped Resource Principal Session Tokens with **JWT** claim-based least-privilege checks, satisfying an internal Oracle security review blocking launch.

Verizon | *Java, Spring Boot, Spring Batch, Docker, Helm, Kubernetes, Jenkins, Argo CD*

Basking Ridge, NJ

Software Engineer Intern

Jun 2024 – Aug 2024

- Cut KPI refresh latency from nightly to hourly across Verizon network metrics by building a **Spring Boot + Spring Batch** ingestion service that normalizes raw data sources into a canonical schema.
- Exposed normalized KPI data to Verizon's network ops and reliability dashboards by designing **REST** endpoints in **Spring Boot** that served as the canonical data source for downstream consumers.

Software Engineer Intern

Jun 2023 – Aug 2023

- Reduced manual release effort on PlatformZero by containerizing services with **Docker/Helm** for **Kubernetes** and configuring **Argo CD** applications to automate **Jenkins**-based releases under a **GitOps** model.
- Caught regressions pre-merge on PlatformZero APIs by writing **JUnit** and integration test suites across the core endpoint surface.

Avis Budget Group | *Kotlin, Android, Python, Tableau*

Parsippany, NJ

Software Engineer Intern

Jun 2022 – Aug 2022

- Eliminated manual environment-switching setup for the QA team by building an internal **Android** tool in **Kotlin** that automated configuration updates across test environments.

Projects

Boston Public Library: Retrieval-Augmented Search | *Python, PostgreSQL, pgvector, OpenAI, Streamlit*

- Built a **RAG** pipeline over 1.2M+ Boston Public Library records using **PostgreSQL** with **pgvector**, enabling semantic search across 5M+ embedded text chunks with 2–5 second query latency.
- Designed chunking and embedding strategies for archival text, balancing chunk size and overlap to preserve context across paragraph boundaries and improve semantic match quality.
- Added LLM-based query expansion and **BM25** reranking with **OpenAI GPT-4o-mini** on top of vector search to handle historical natural language queries that pure keyword matching missed.

Otto – AI-Powered Used Car Deal Finder | *Python, FastAPI, PostgreSQL, Redis, Claude API, Docker*

- Built a multi-worker **Python** pipeline that sweeps Facebook Marketplace nationwide via a headless **Chromium** pool, ingesting and deduping listings across 5 independently scalable workers.
- Designed an enrichment and scoring chain combining **NHTSA** VIN decoding, market-comp aggregation, and a vision + text pass via the **Claude API** to produce a 0–100 deal score.

Education

Boston University

Boston, MA

Master of Science in Computer Science & Bachelor of Arts in Computer Science

May 2025

- **GPA:** 3.80/4.00 (MS), 3.76/4.00 (BA)
- **Relevant Coursework:** Distributed Systems, Database Systems, Machine Learning, Computer Networks, Cloud Computing, Operating Systems
- **Leadership:** Teaching Assistant (Database Systems), Course Assistant (Comp. Systems)